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LAB REPORT 8

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Submitted By:

Name	Student ID
Naheda Islam Ela	2024-3-60-080
Abdullah Al Junaid	2024-3-60-082
Jumana Haider	2024-3-60-594

Submitted To:

Dr. Md. Habibur Rahman

Adjunct faculty

Department of Computer Science and Engineering

East West University

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Experiment Title:

To study the transfer characteristics of a MOSFET and to use it as a switch.

Objectives

1. To understand the basic structure and the working principle of a MOSFET, and to see how the drain current changes when the gate voltage is changed.
2. To study the transfer characteristics of a MOSFET, which show the relation between the gate to source voltage and the drain current, and to find out the threshold voltage from the recorded data.
3. To learn how a MOSFET can be used as a simple switch to turn a load, such as an LED, on and off, and to compare its switching behavior in the low state and in the high state.

Theory

A MOSFET, which stands for Metal Oxide Semiconductor Field Effect Transistor, is a three terminal device that is widely used in modern electronic circuits. Unlike a BJT, which is controlled by current, a MOSFET is controlled by voltage. This means that only a very small current is needed at the input to control a much larger current at the output. Because of this property, a MOSFET is very useful in digital circuits, switching circuits and power control applications. The three terminals of a MOSFET are given below.

Gate (G): This is the input terminal of the device. The voltage that is applied at this terminal decides whether the MOSFET stays off or turns on.

Drain (D): This is the terminal through which current enters the device when it is turned on.

Source (S): This is the terminal through which current leaves the device. It is usually connected to the ground or to the common line of the circuit.

In this experiment, the MOSFET 2N7000 was used, which is an N channel enhancement type MOSFET. In an enhancement type MOSFET, the device stays off as long as the gate to source voltage, V_{GS} , is below a certain value. This value is called the threshold voltage, and it is usually written as V_{th} . Once V_{GS} becomes greater than V_{th} , a thin channel forms between the drain and the source, and current is allowed to flow through the device. As V_{GS} is increased further above the threshold voltage, the drain current, I_D , also keeps increasing.

Transfer Characteristics

The transfer characteristics of a MOSFET show the relation between the gate to source voltage, V_{GS} , and the drain current, I_D , while the drain to source voltage, V_{DS} , is kept fixed at a constant value. When V_{GS} is small and is below the threshold voltage, the drain current stays close to zero, because the MOSFET remains in the cut off region and no channel has formed yet. Once V_{GS} crosses the threshold voltage, the drain current starts to rise. At first the rise is slow, but as V_{GS} is increased further, the drain current increases much faster. This general behavior can be represented using a simple curve, which is shown in Figure 1 below.

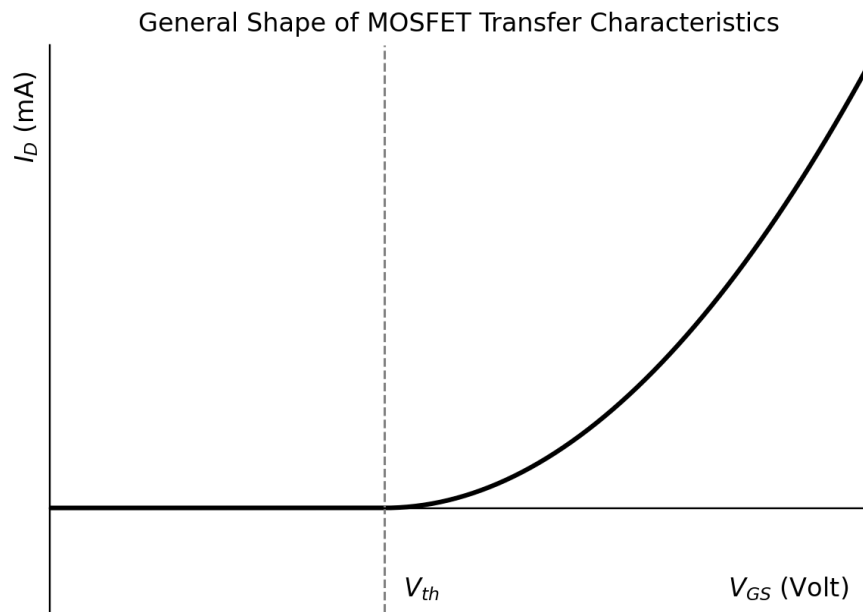


Figure 1: General shape of the transfer characteristics of a MOSFET, showing the threshold voltage V_{th}

MOSFET as a Switch

A MOSFET can also be used as an electronic switch, in a way that is similar to a normal mechanical switch, but without any moving parts. When the gate voltage is low, that is below the threshold voltage, the MOSFET stays in the cut off region and behaves like an open switch, so almost no current flows through the load that is connected to it. When the gate voltage is high, that is above the threshold voltage, the MOSFET enters the ohmic region, sometimes called the triode region, and behaves like a closed switch. In this state, current is allowed to flow through the load, and only a very small voltage drop is left across the drain and source terminals of the MOSFET.

In this experiment, an LED was connected as the load so that the switching action of the MOSFET could be seen directly. When the input voltage, V_{in} , was kept low, the LED stayed off. When V_{in}

was made high, the LED turned on. This clearly shows that a MOSFET can be used to control a load using only a small control voltage applied at the gate terminal.

Apparatus Required

1. MOSFET: 2N7000
2. Resistors: 10k Ohm, 4.7k Ohm
3. LED
4. Breadboard
5. Connecting wires
6. Multimeter
7. Power supply
8. Ammeter

Working Diagrams

i) Transfer Characteristics

To study the transfer characteristics, the gate to source voltage, V_{GS} , was changed step by step using a variable gate supply, while the drain to source voltage, V_{DS} , was kept fixed at 15 volt using a separate supply. For every value of V_{GS} , the drain current, I_D , was noted down using the ammeter connected in the drain path. The circuit that was used for this part of the experiment is shown in Figure 2.

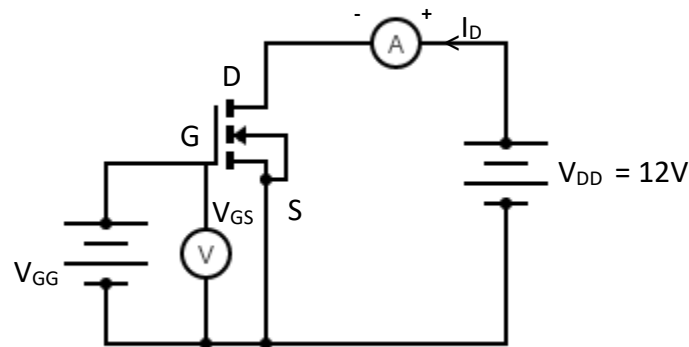


Figure 2: Circuit diagram used to study the transfer characteristics of a MOSFET

ii) MOSFET as a Switch

To check the switching behavior of the MOSFET, the gate was connected to an input voltage, V_{in} , through a gate resistor, R_G , of 10k Ohm. An LED, along with a load resistor, R_D , of 4.7k Ohm, was connected at the drain terminal, and a supply voltage, V_{DD} , of 12 volt was used. When V_{in} was low, the LED stayed off, and when V_{in} was high, the LED turned on. The circuit that was used for this part of the experiment is shown in Figure 3.

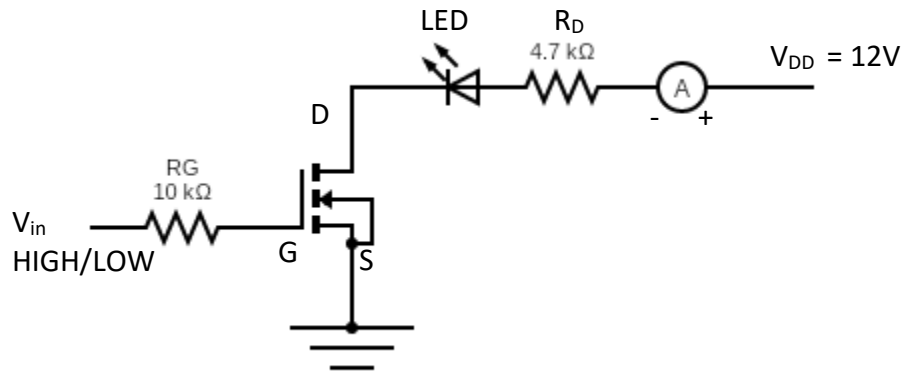


Figure 3: Circuit diagram used to study the MOSFET as a switch

Data Table

Transfer Characteristics

No. of Obs	VGS (Volt)	ID (mA)	VDS (Volt)
1	0	0	15
2	0.5	0	15
3	0.81	0.6	15
4	0.97	1	15
5	1	1.4	15
6	1.07	1.8	15
7	1.1	2.2	15
8	1.2	4	15
9	1.24	4.45	15
10	1.27	5.8	15

No. of Obs	VGS (Volt)	ID (mA)	VDS (Volt)
11	1.3	6.8	15
12	1.32	7.6	15
13	1.34	8.45	15
14	1.365	9.85	15
15	1.37	10	15

MOSFET as a Switch

No. of Obs	Vin	State of the LED	Load Current (mA)	Load Voltage (Volt)	VDS (Volt)
1	LOW	OFF	0.00	0.372	10.86
2	HIGH	ON	2.3	11.84	0.1

The recorded values of VGS and ID from the transfer characteristics data table were also plotted on a graph, so that the shape of the curve could be seen clearly. This graph is shown in Figure 4.

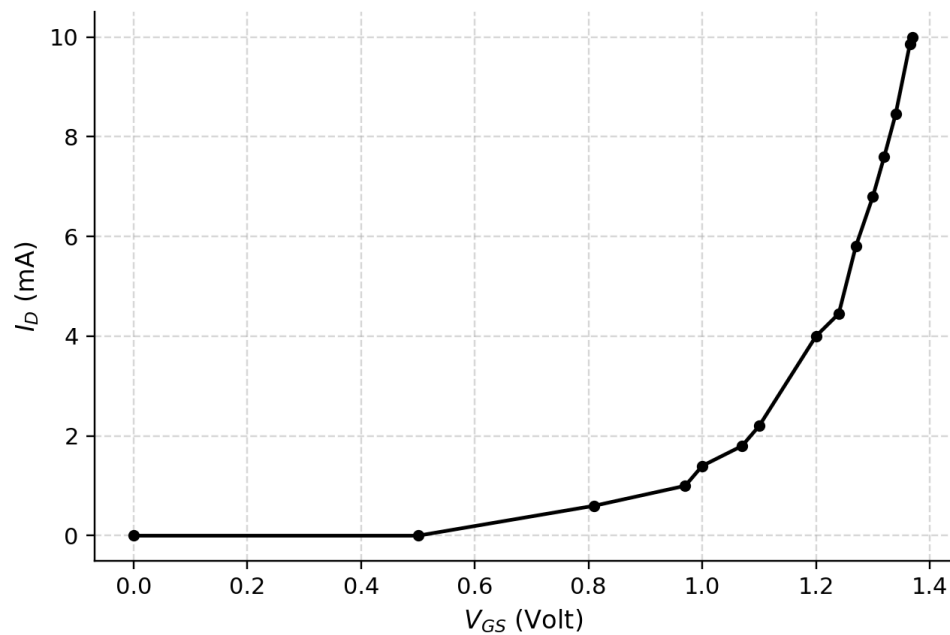


Figure 4: Transfer characteristics curve of the MOSFET 2N7000

Discussion

In this experiment, we studied the transfer characteristics of a MOSFET and also used the same MOSFET as a switch. From the first data table, it can be seen that when the gate to source voltage, V_{GS} , was zero or very small, the drain current, I_D , stayed at zero. This shows that the MOSFET was in the cut off region and no channel had formed between the drain and the source yet. As V_{GS} was increased slowly, the drain current started to rise once V_{GS} crossed a value close to 0.8 volt to 1 volt. This value can be considered as the threshold voltage of this particular MOSFET. After this point, the drain current increased faster as V_{GS} was increased further, which matches the general shape shown earlier in Figure 1 and also matches the plotted graph in Figure 4.

From the second data table, where the MOSFET was used as a switch, it can be seen that when V_{in} was low, the LED remained off, the load current was almost zero, and the drain to source voltage, V_{DS} , was high, close to the supply voltage. This shows that the MOSFET was behaving like an open switch. On the other hand, when V_{in} was high, the LED turned on, a load current of about 2.3 mA was recorded, and the drain to source voltage dropped to a very small value close to zero. This shows that the MOSFET was behaving like a closed switch, and almost all of the supply voltage was appearing across the load instead of across the MOSFET.

From this experiment, it can be understood that a MOSFET is able to control a fairly large output current using only a small input voltage applied at the gate, without drawing almost any input current at all. This property makes the MOSFET very useful for digital switching, power control circuits and many modern electronic devices. Some common uses of a MOSFET switch are in power supplies, motor control circuits, LED lighting control and digital logic circuits.

For future improvement, this experiment can be extended by also testing a P channel MOSFET and comparing its behavior with the N channel MOSFET that was used here. It would also be useful to build more advanced circuits, such as a MOSFET based H bridge, which is commonly used to control the direction of a motor, so that a deeper understanding of practical MOSFET applications can be developed.